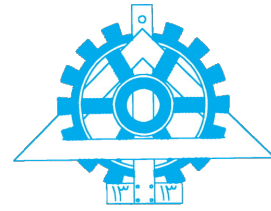




University of Tehran
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Introduction to Data Science

Assignment 1

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Introduction

This report outlines the findings and visualizations for Data Science Assignment 1. The project encompasses three primary practical tasks: statistical sampling and inference using 2D Gaussian distributions, storytelling through interactive Tableau dashboards for Global Tuberculosis data, and the visual improvement of a financial Power BI dashboard. The programmatic implementation for data processing and mathematical modeling can be found in the accompanying Jupyter Notebook.

Task 1: Sampling and Inference

2.1 1. Define and Visualize the Distribution

Figure 1 displays the probability density function (PDF) of a standard 2D Gaussian distribution centered at the origin $(0,0)$. The color intensity represents the probability density, with the darkest red region indicating the highest probability at the mean. The color fades outward in a perfectly uniform gradient as the probability density approaches zero at the tails of the distribution.

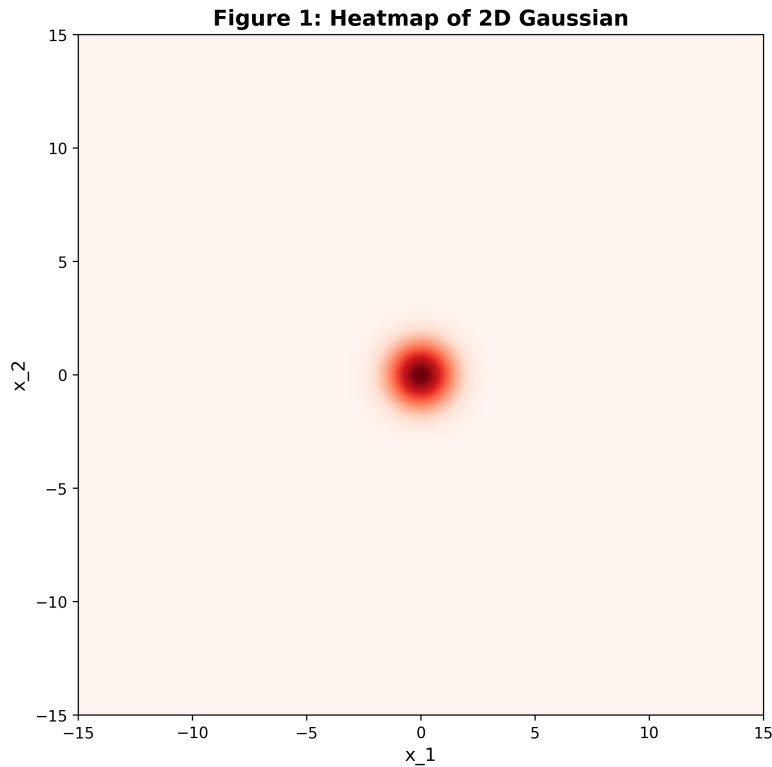


Figure 1: Heatmap of the 2D Gaussian target distribution ($\mu = [0, 0]$, $\Sigma = I$)

Theoretical Questions

1. Why is the distribution symmetric?

The distribution is symmetric because the covariance matrix is the Identity matrix (I). This means the variance along both the x_1 and x_2 axes is identically equal to 1, and the variables are completely independent (their covariance is 0). Because the spread is equal in all directions and there is no correlation skewing the data, the probability density depends purely on the Euclidean distance from the mean, resulting in perfect circular (radial) symmetry.

2. What role does the covariance matrix play in the shape of the distribution?

The covariance matrix completely dictates the spatial spread, shape, and orientation of the Gaussian distribution:

- The **diagonal elements (variances)** control the width or spread of the distribution along each individual axis.
- The **off-diagonal elements (covariances)** control the linear correlation between the variables, which determines the "tilt" or rotation of the distribution.

In our specific case, because the off-diagonal elements are exactly zero and the diagonal elements are equal, the distribution has no tilt and forms a perfect circle rather than an elongated

ellipse.

2.2 2. Sampling Methods

In this section, we generate 1000 samples from the target distribution using two distinct methods: direct sampling via NumPy's built-in multivariate normal generator, and Langevin Dynamics sampling.

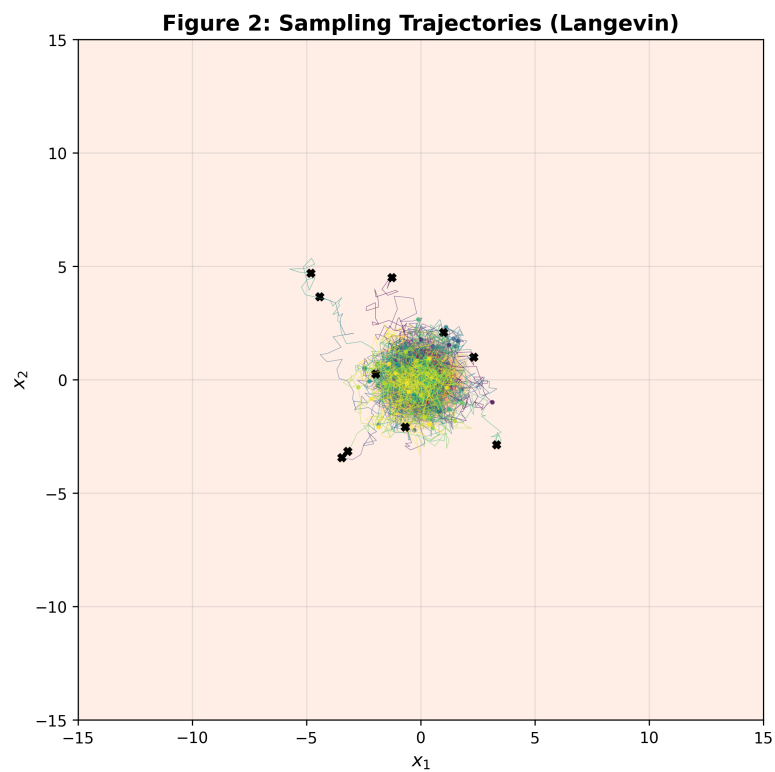


Figure 2: Sampling Trajectories of 10 particles using Langevin Dynamics

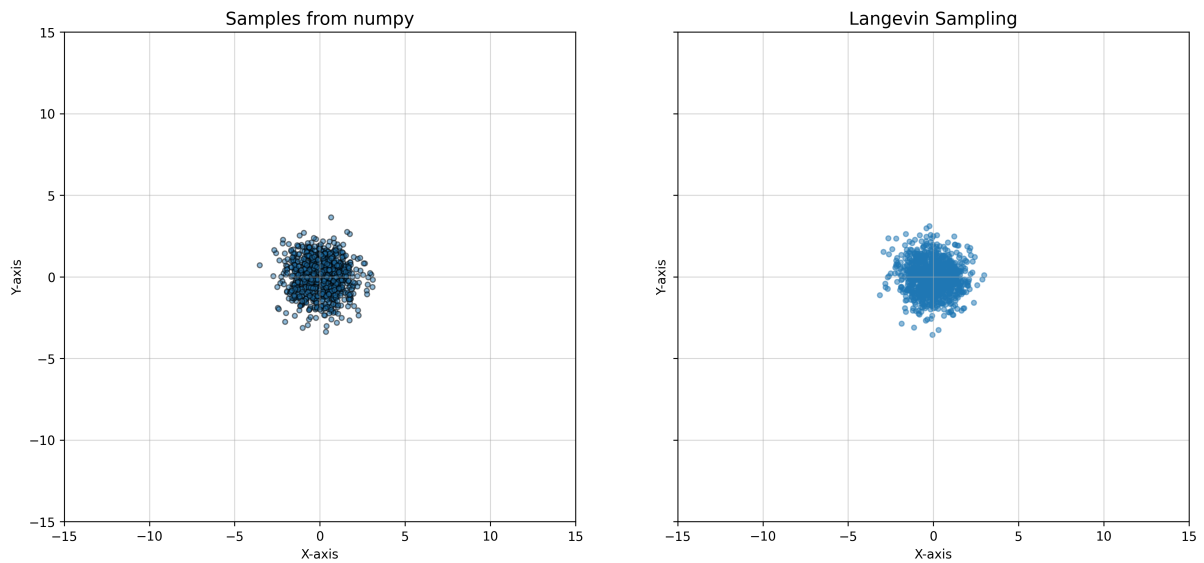


Figure 3: Comparison of Final Sampled Points: Direct Sampling (left) vs. Langevin Dynamics (right)

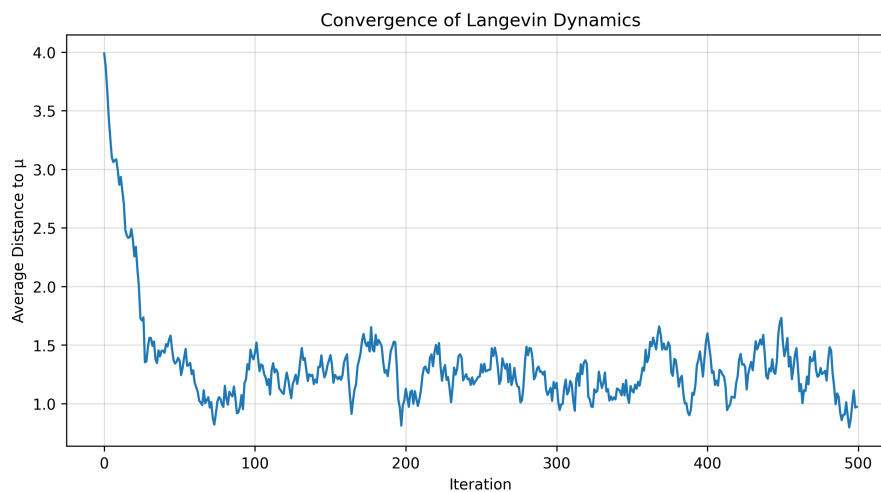


Figure 4: Convergence Analysis: Average distance of samples from the origin over iterations

Questions

1. How does changing the step size affect convergence speed and stability of the sampler?

The step size (ϵ) controls the balance between speed and stability. A larger step size allows particles to take larger strides toward the high-density regions, converging faster, but it risks overshooting the target distribution, leading to instability. Conversely, a smaller step size ensures stability but requires significantly more iterations to reach the stationary distribution.

2. How many iterations are required for the sampler to reach a stable distribution for your chosen step size? Explain how you determined this.

Based on the convergence plot (Figure 4), the sampler reaches a stable distribution after approximately 30 iterations. This was determined by observing the average distance of the particles from the origin. The distance drops rapidly during the initial phase and then levels off around iteration 30, where it begins to steadily oscillate.

3. Compare the scatter plot with the density contours and explain why more samples appear near the origin.

The density contours represent the probability density function, which peaks at the origin (0,0) for our standard Gaussian. The samples cluster densely around this central point because the score function ($\nabla_x \log p(x)$) actively directs the particles toward regions of higher probability, pulling them inward from the edges.

4. If you remove the noise term from the update rule, what behavior do you expect and why?

If the random noise term ($\sqrt{2\epsilon}z_t$) is removed, the update rule becomes purely deterministic, effectively turning into gradient ascent. Without the noise to disperse the particles, every particle would climb the probability gradient and permanently collapse exactly onto the origin (the mode). It would find the maximum probability point but fail to sample the variance of the distribution.

5. Do all trajectories converge to similar regions of the space? What does this indicate about the sampler?

Yes, as shown in the trajectory plot, all particle trajectories converge toward the central region around the origin, but they continue to fluctuate within that area rather than converging to a single point. This indicates that the sampler is working correctly: it successfully navigates to the high-probability region while maintaining the natural variance expected from the target Gaussian distribution.

2.3 3. Inference from Samples

Based on the programmatic evaluation, we calculated the estimated means and variances for both sampling methods and compared them against the true target parameters ($\mu = [0, 0]$, $\text{Var} = [1, 1]$). The results of the 1000 generated samples are summarized in Table 1.

Metric	NumPy Direct Sampling	Langevin Dynamics
Estimated Mean (x_1, x_2)	[0.0548, 0.0109]	$[-0.0582, -0.0358]$
Estimated Variance (x_1, x_2)	[1.0085, 1.0462]	[0.9567, 0.9897]
Absolute Mean Error	[0.0548, 0.0109]	[0.0582, 0.0358]
Absolute Variance Error	[0.0085, 0.0462]	[0.0433, 0.0103]

Table 1: Inference parameters and absolute errors for both sampling methods.

As seen in the results, Direct Sampling produced very low absolute errors because it draws exact, independent samples. Langevin Dynamics also approximated the parameters well, but showed slightly higher errors. This happens because the computer simulation must estimate the continuous paths using discrete, straight-line jumps, which introduces a small mathematical bias.

Questions

1. Why can the Langevin estimates still be biased even when the final samples come from independent chains?

The bias in Langevin dynamics is fundamentally caused by the step size (discretization error), not by a lack of samples. In a perfect mathematical model, the particles move in perfectly smooth, continuous curves. However, the computer forces the particles to take distinct, straight-line jumps. Because these jumps are straight, they slightly overshoot the true, curved shape of the distribution. Even if we run multiple completely independent chains (particles) to reduce random noise, every single chain is still forced to use this same flawed "straight-line" approximation. Therefore, their final results will always average out to a slightly distorted distribution.

2. How do the number of iterations T and the step size ϵ affect the accuracy of the Langevin samples?

The accuracy relies on a trade-off between these two parameters:

- **Number of iterations (T):** This determines how long the simulation runs. A larger T is necessary to give the particles enough time to move away from their random starting locations (overcoming "burn-in") and fully explore the target distribution.
- **Step size (ϵ):** This controls the physical length of each jump. A smaller ϵ makes the straight-line jumps tiny, which allows them to trace the true curves of the distribution

much more accurately (lower bias). However, taking tiny steps means the particles move very slowly across the space, so you are forced to use a much larger T to give them enough time to explore. Conversely, a larger step size lets them explore faster, but increases the error because large jumps heavily overshoot the curves.

2.4 4. Compare the Sampling Methods

A comprehensive analytical and numerical comparison between NumPy's Direct Sampling and Langevin Dynamics was conducted in the previous sections. Specifically, the visual tracking of global particle trajectories can be referred to in Section 1.2 (Figure 2), while the complete numerical error analysis, including estimated means, variances, and precise absolute errors, is fully detailed in the inference table of Section 1.3 (Table 1). This section builds upon those findings by focusing on coordinate-level stability and visual density matching.

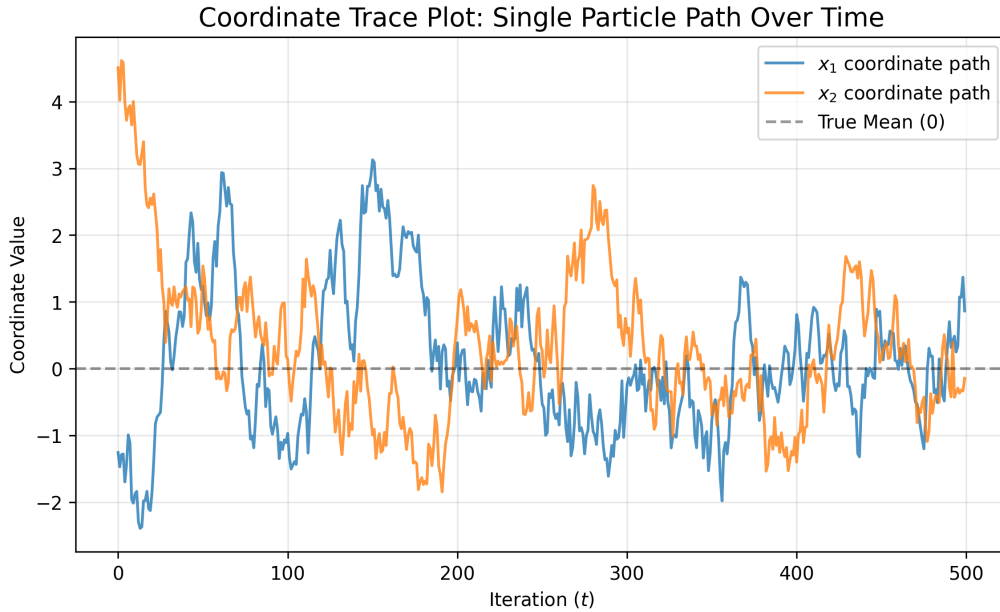


Figure 5: Coordinate Trace Plot: Evolution of x_1 and x_2 for a single particle over 500 iterations.

Coordinate Convergence Analysis

The coordinate trace plot (Figure 5) maps the progression of individual coordinate dimensions over time. The tracking shows a highly dynamic initial phase where the particle positions fluctuate widely as they are pulled toward the high-density regions. By approximately iteration 30, both the x_1 and x_2 coordinates stabilize near the true mean of 0. Beyond this point, the

paths show steady, bounded oscillations. This indicates that the Markov chain has successfully reached its stationary distribution, with the injected random noise maintaining the target variance without causing the system to drift.

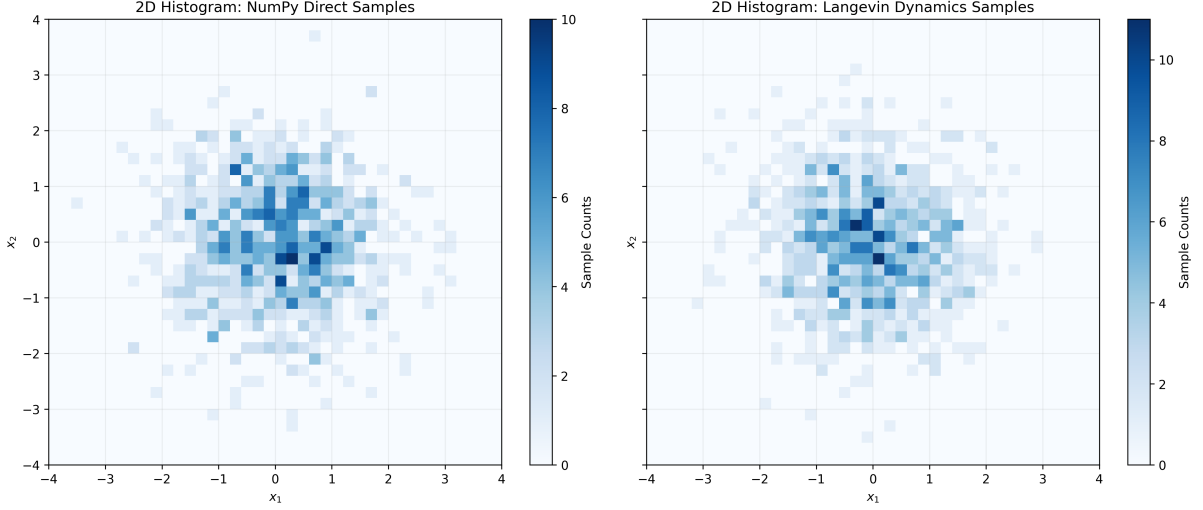


Figure 6: 2D Histogram Density Comparison: NumPy Direct Sampling (left) vs. Langevin Dynamics (right).

Density Comparison and Conclusion

The side-by-side 2D histograms (Figure 6) divide the space into discrete bins to show the exact density concentration of the generated samples.

- **Direct Sampling:** Displays a perfectly concentric, radially symmetric density pattern that matches a standard 2D Gaussian distribution flawlessly. This is because the samples are fully independent and drawn directly from the true distribution.
- **Langevin Dynamics:** Strongly mirrors this central concentration, proving that the score-driven gradient vector successfully pulls the particle population into the correct density shape.

Minor local variations or asymmetric textures visible in the Langevin histogram are standard artifacts of running a finite number of iterations ($T = 500$) with a discrete step size ($\epsilon = 0.1$). This visualization confirms that despite minor numerical discretization errors, Langevin dynamics successfully explores and maps the true underlying probability density of the target space.

Task 2: Tableau Dashboard Storytelling

This section explains how we prepared our data, built the interactive dashboards, and created a clear story to analyze the Global Tuberculosis (TB) data.

3.1 Data Integration & Preparation

To add region and location details to our main disease data, we combined three files in Tableau:

- `TB_Burden_Country.csv`: Contains the main country-level TB data over time.
- `Region.csv`: Contains the official WHO regional groups.
- `Coordinate.csv`: Provides the Latitude and Longitude for mapping.

We used a **Left Outer Join** to connect the `Region` and `Coordinate` tables to the main `TB_Burden` table. We linked them using the **"Country or territory name"** and **"ISO numeric code"**. We chose a Left Join to make sure we keep all the region and map data, even if some countries were missing TB data. This keeps our maps and regional groups working perfectly.

To keep our analysis accurate, we handled missing values (nulls) very carefully:

- **Avoiding Fake Zeros**: We did not replace missing values with zeros (like using the `ZN` function). Changing missing data to absolute zeros creates wrong statistics. For example, if we put zero for missing HIV data, it would incorrectly lower the averages and give us wrong results.
- **Filtering Out Nulls**: Instead, we removed null records using a `Non-null values` filter. This makes sure our charts, averages, and trend lines only show real, reported data. For the heatmap, we simply showed **"N/A"** to show that data was not reported.

3.2 Dashboard Interactions

To make the dashboards easy to explore and analyze, we added several interactive filtering actions across the charts:

- **Dashboard 1 (The Global Footprint)**: We used a two-way filter between the Regional Treemap and the World Map. Selecting a region on the Treemap filters the map to show

only those countries. Conversely, clicking a country on the map highlights its region in the Treemap. We also included a **Year Slider** so users can move back and forth through time to watch how the disease spreads globally.

- **Dashboard 2 (The Syndemic):** Selecting a specific region's bar in the "Co-Infection Burden" chart automatically filters the "Mortality Accelerator" scatter plot. This allows the user to temporarily hide other regions and focus strictly on the relationship between HIV and TB for the selected area.
- **Dashboard 3 (System Efficacy):** We linked three charts together using filter actions. Clicking a square in the "Diagnostic Blindspots" heatmap isolates that country as a single dot on the "Detection vs. Survival" scatter plot, and vice versa. Furthermore, clicking any country in the "Accountability List" updates the other two charts to focus solely on it. To make the analysis easier and more actionable, we pre-filtered the Accountability List to only show the **Top 15 worst-performing countries**.

3.3 Custom Calculated Fields

To build the scatter plots, median lines, and dynamic colors, we created several custom calculated fields in Tableau. The core logic for these custom variables is structured as follows:

- **Estimated incidence of TB (all forms):** Represents the absolute estimated number of new and relapse Tuberculosis cases diagnosed within a given year.

```
(([Estimated total population number]
  * [Estimated incidence (all forms) per 100 000 population])
 / 100000
```

- **Estimated mortality of TB (all forms) per 100k:** Specifies the standardized rate of TB-related deaths per 100,000 individuals in the population.

```
[Estimated mortality of TB cases (all forms, excluding HIV)
 per 100 000 population]
+ [Estimated mortality of TB cases who are HIV-positive,
 per 100 000 population]
```

- **Total Deaths:** Calculates the total volume of mortality by converting the standardized absolute death counts.

```
[Estimated number of deaths from TB  
  (all forms, excluding HIV)]  
+ [Estimated number of deaths from TB in people  
  who are HIV-positive]
```

- Global Median Detection: Calculates the middle point for case detection to divide the scatter plot into distinct areas.

```
{FIXED : MEDIAN(  
  IF NOT ISNULL([Case detection rate (all forms), percent])  
  THEN [Case detection rate (all forms), percent]  
  END  
)}
```

- Global Median Mortality: Calculates the middle point for the mortality rate across all countries for a given year.

```
{FIXED : MEDIAN(  
  IF NOT ISNULL([Estimated mortality of TB (all forms)  
    per 100 000 population])  
  THEN [Estimated mortality of TB (all forms)  
    per 100 000 population]  
  END  
)}
```

- Mortality Rate per 100k: Standardizes the death count based on population size for fair comparisons.

```
[Estimated mortality of TB cases (all forms, excluding HIV)  
  per 100 000 population]  
+ [Estimated mortality of TB cases who are HIV-positive,  
  per 100 000 population]
```

- Quadrant Color: A conditional statement that assigns colors to countries based on where they fall compared to the global medians.

```

IF AVG([Case detection rate (all forms), percent])
  < ATTR([Global Median Detection])
AND AVG([Estimated mortality of TB (all forms) per 100 000 population])
  >= ATTR([Global Median Mortality])
THEN 'System Failure (Top Left)'

ELSEIF AVG([Case detection rate (all forms), percent])
  >= ATTR([Global Median Detection])
AND AVG([Estimated mortality of TB (all forms) per 100 000 population])
  >= ATTR([Global Median Mortality])
THEN 'Clinical Failure (Top Right)'

ELSEIF AVG([Case detection rate (all forms), percent])
  >= ATTR([Global Median Detection])
AND AVG([Estimated mortality of TB (all forms) per 100 000 population])
  < ATTR([Global Median Mortality])
THEN 'Success Zone (Bottom Right)'

ELSE 'Low Detection & Low Mortality (Bottom Left)'
END

```

postseason

3.4 The 6-Stage Story

We built a 6-step Tableau Story to show our findings clearly, from a global view down to an action plan.

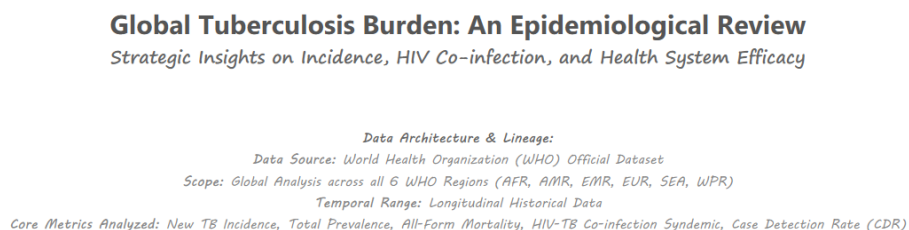


Figure 7: Executive Summary and Data Lineage Overview.

1. Stage 1: Executive Summary

- *Focus*: A clean page showing data sources and research architecture.
- *Narrative*: This stage establishes the empirical foundation of the study using official, longitudinal datasets from the World Health Organization (WHO). It delineates the architectural lineage across all six major WHO epidemiological regions (AFR, AMR, EMR, EUR, SEA, WPR). By clearly mapping out the core metrics—such as incidence, prevalence, all-form mortality, and HIV co-infection—before diving into the analytics, this stage establishes data transparency, ensures scientific credibility, and sets a structured framework for the entire analytical narrative.

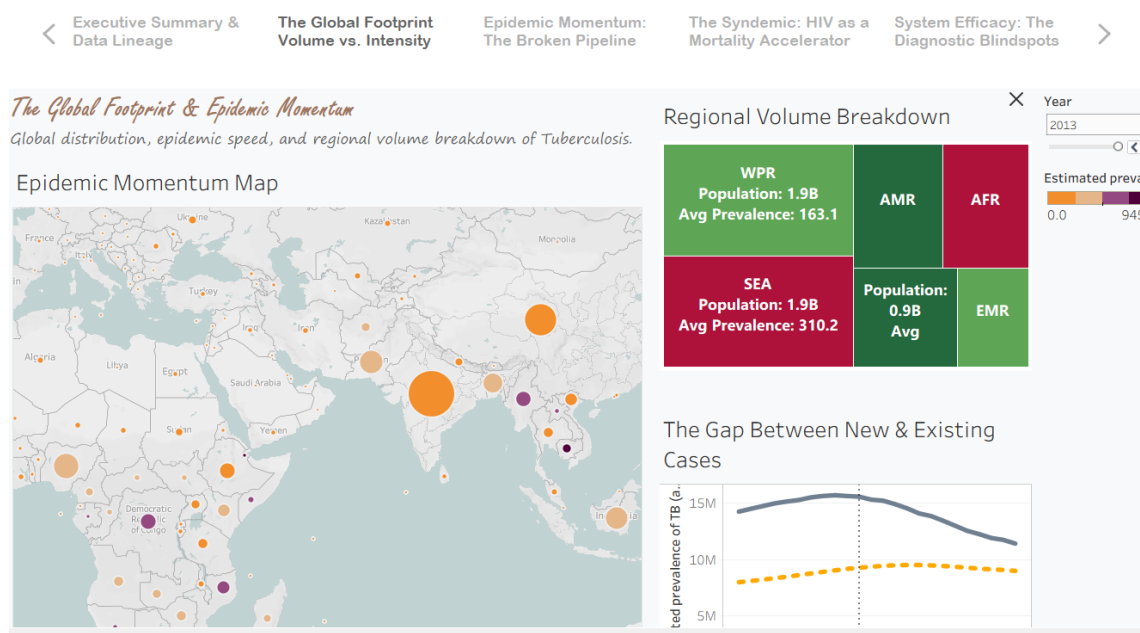


Figure 8: The Global Footprint Dashboard: Volume vs. Intensity Analysis.

2. Stage 2: The Global Footprint

- *Focus*: Bubble Map and Regional Prevalence Treemap side-by-side.
- *Narrative*: This stage unveils a critical epidemiological dichotomy between absolute case volume and infection intensity. High-population nations in the South-East Asia (SEA) and Western Pacific (WPR) regions, specifically giants like India and China, exhibit massive absolute case volumes represented by the large bubble sizes, driven by sheer population scale. Conversely, the African region (AFR) displays a disproportionately severe epidemiological intensity; despite lower absolute populations, its darker shading signifies extremely high per-capita prevalence rates. This highlights that while Asian nations require heavy infrastructural scaling, African nations demand urgent, targeted clinical interventions to lower infection density.

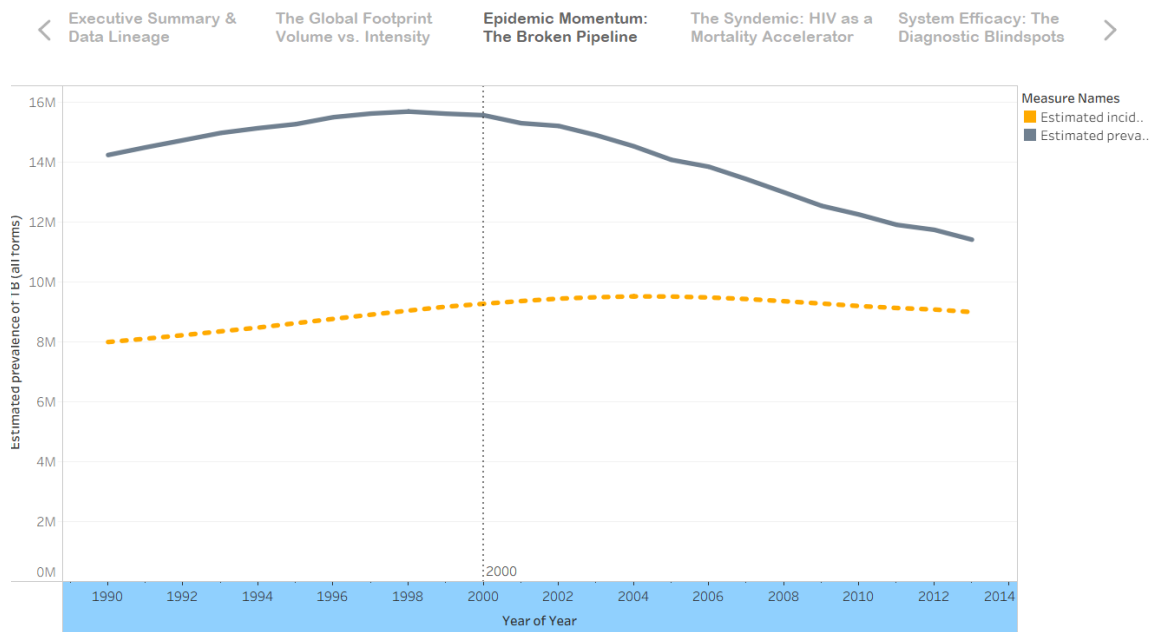


Figure 9: Epidemic Momentum: The Broken Pipeline (Prevalence vs. Incidence Trends).

3. Stage 3: The Broken Pipeline

- *Focus:* Dual-Axis Line Chart tracking Prevalence vs. Incidence trends over time.
- *Narrative:* This chart traces the historical trajectory of the epidemic, highlighting a vital paradigm shift marked by the vertical milestone line at the year 2000. Prior to this era, both prevalence and incidence were climbing rapidly. Post-2000, due to substantial advancements in medical technology, global sanitation, and targeted public health campaigns, the two curves begin to converge as total prevalence drops significantly. Concurrently, the slope of new infections (incidence) has flattened and started a gradual decline; however, it maintains a stubbornly persistent baseline because of explosive global population growth, signaling that although control mechanisms are working, transmission pipelines are not yet fully broken.

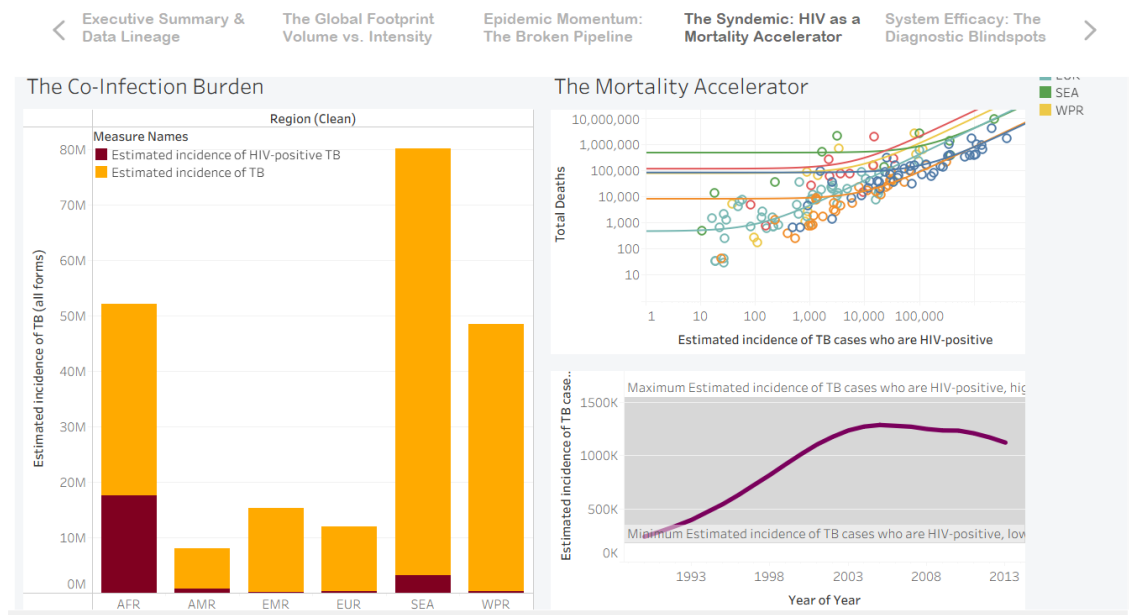


Figure 10: The Syndemic: HIV Co-infection as a Mortality Accelerator.

4. Stage 4: HIV as an Accelerator

- *Focus:* Logarithmic Scatter Plot with regional trend lines coupled with a Co-Infection Burden Bar Chart.
- *Narrative:* This stage investigates the deadly intersection of the TB-HIV syndemic, demonstrating how HIV acts as a catastrophic accelerator of human mortality. The log-scale scatter plot illustrates a tight, positive correlation between the incidence of HIV-positive TB cases and total deaths, proving that co-infection exponentially decreases patient survival rates. Regionally, the bar chart exposes the African region (AFR) as the epicenter of this dual crisis, holding an overwhelmingly large share of HIV-positive TB cases compared to other regions like South-East Asia (SEA). This demands an integrated clinical approach where HIV antiretroviral therapies and TB antibiotics are co-administered simultaneously.

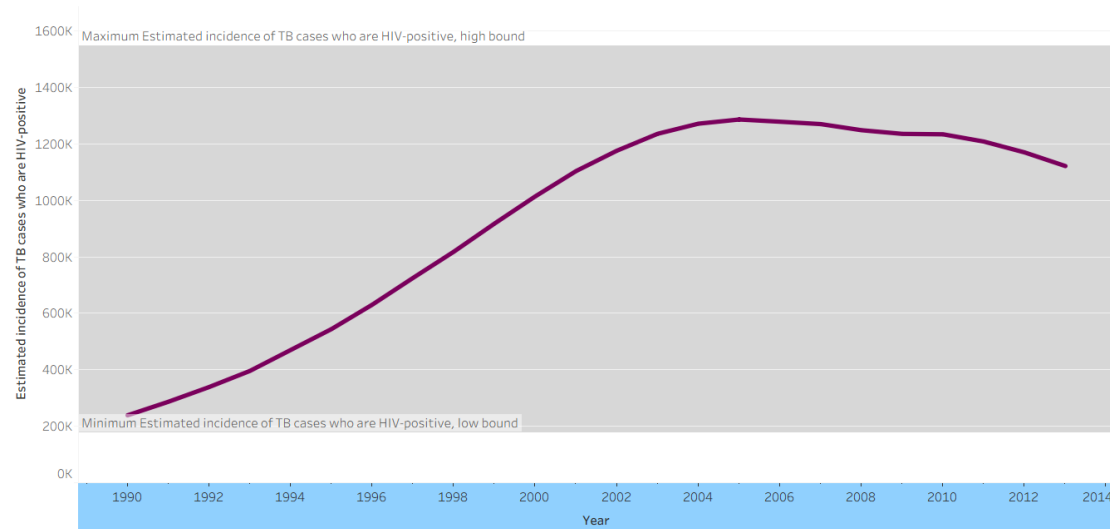


Figure 11: Epidemiological Variance: Longitudinal Decline of the Estimation Confidence Band.

5. Stage 5: Estimation Confidence Band

- *Focus*: Empirical Uncertainty Bounds and Epidemiological Variance.
- *Narrative*: This visualization tracks the statistical confidence intervals of global TB estimates, exposing a profound structural turning point around the year 2006. Prior to this era, the wider confidence bands reflected substantial data volatility and lower tracking precision across developing healthcare infrastructures. Post-2006, the prominent downward trajectory of the estimation band signifies a major expansion in data precision. This drop is driven by two catalyst factors: the digitization of global surveillance systems which standardized country-level reporting, and the emergence of revolutionary molecular diagnostic technologies. By replacing mathematical projections with empirical, active case tracking, the statistical margin of error collapsed, reinforcing the scientific reliability of global intervention frameworks.

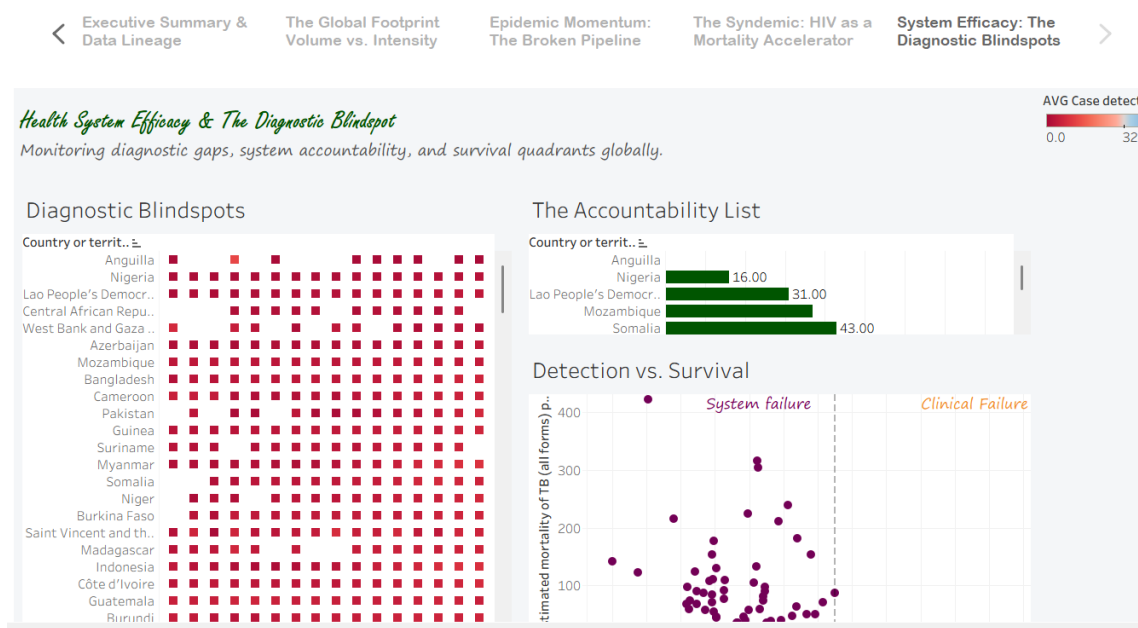


Figure 12: System Efficacy: Heatmap of Global Diagnostic Blindspots.

6. Stage 6: Finding the Blindspots

- *Focus*: Longitudinal Heatmap of case detection rates.
- *Narrative*: This stage shifts focus from clinical severity to healthcare delivery infrastructure by mapping structural failures via a diagnostic heatmap. The persistent red indicators over consecutive years flag critical "blindspots" where case detection rates routinely plummet below acceptable thresholds in countries like Nigeria, Mozambique, and Somalia. This visualization proves that the primary bottleneck in these healthcare systems is a systematic failure to screen, diagnose, and register active cases within communities, rather than a mere shortage of pharmaceutical treatments. Without active case-finding and diagnostic expansion, hidden reservoirs of the disease will continue to drive community transmission.

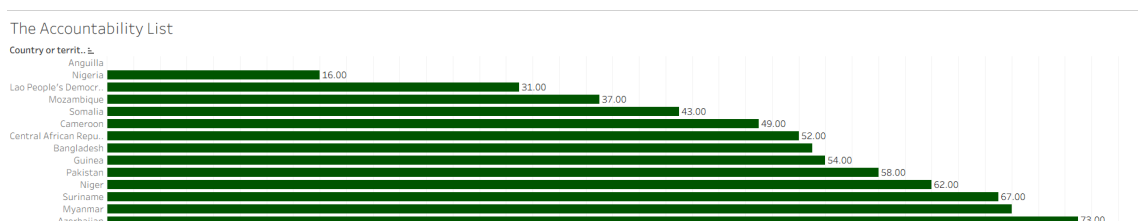


Figure 13: Action Plan: Detection vs. Survival Quadrant Scatter Plot and Priority List.

7. Stage 7: Action Plan

- *Focus*: Detection vs. Survival Quadrant Scatter Plot and a filtered Priority Accountability List.
- *Narrative*: This final stage synthesizes previous insights into an actionable, data-driven governance framework by plotting nations along a "Detection vs. Survival" quadrant scatter plot. It isolates the most vulnerable countries within the critical "System Failure" quadrant—characterized by dangerously low case detection rates combined with high mortality. By coupling this with a prioritized "Accountability List" of the top 15 worst-performing nations (such as Nigeria and Somalia), the dashboard transcends descriptive analytics to provide a strategic blueprint for international aid. It recommends that global health funding must be reallocated away from generic programs and directed explicitly toward upgrading laboratory networks, molecular testing infrastructure, and community-based surveillance in these high-priority zones.

3.5 Visual Design Principles

To make the dashboard easy to read, we used standard design rules:

- **Gestalt Principles (Grouping)**: We used a clean grid layout with white cards on a light gray background. This helps the user easily see which charts belong together. We also removed messy gridlines on line charts to make them easier to follow.
- **Colors and Sizes**: We used a calm color palette, keeping dark Purple only for danger zones so it catches the eye immediately. We also used bubble sizes to clearly show the number of cases.